

Timo Probst

Research Assistant
Chair of Computer Graphics, Department for Computer Science
University of Freiburg, Germany

✉ probstt@informatik.uni-freiburg.de  Personal Website  LinkedIn  Google Scholar

EDUCATION

Ph.D. Student <i>Chair of Computer Graphics, Department of Computer Science</i> <i>Supervisor: Matthias Teschner</i>	University of Freiburg, Germany 2021 – ...
M.Sc. in Computer Science <i>Graduated with honors.</i> <i>Thesis title: Global Contact Handling and Dry Friction for Particle-based Rigid Bodies.</i>	University of Freiburg, Germany 2019 – 2021
B.Sc. in Computer Science <i>Thesis title: Matrix-free PPE Solver for SPH Fluids</i>	University of Freiburg, Germany 2015 – 2019

EMPLOYMENT

Research Assistant <i>Chair of Computer Graphics, Department of Computer Science</i>	University of Freiburg, Germany 2021 – ...
Student Assistant <i>Chair of Computer Graphics, Department of Computer Science</i>	University of Freiburg, Germany 2019 – 2021
Tutor <i>Python for Biologists, Faculty of Biology</i>	University of Freiburg, Germany 2019

PUBLICATIONS

- [1] Timo Probst and Matthias Teschner. “Unified Pressure, Surface Tension and Friction for SPH Fluids”. In: *ACM Trans. Graph.* 44.1 (Dec. 2024). ISSN: 0730-0301. DOI: 10.1145/3708034.
- [2] Timo Probst and Matthias Teschner. “Monolithic Friction and Contact Handling for Rigid Bodies and Fluids Using SPH”. In: *Computer Graphics Forum* 42.1 (2023), pp. 155–179. DOI: 10.1111/cgf.14727.

INVITED TALKS

SIGGRAPH 2025 <i>Unified Pressure, Surface Tension and Friction for SPH Fluids</i>	Vancouver, Canada August 2025
FIFTY2 <i>Simulating Water Droplets with SPH</i> <i>Friction and Contact Handling for Rigid Bodies and Fluids Using SPH</i>	Freiburg, Germany December 2024 March 2023
Eurographics 2023 <i>Monolithic Friction and Contact Handling for Rigid Bodies and Fluids Using SPH</i>	Saarbrücken, Germany May 2023

REVIEWS

ACM SIGGRAPH	2025, 2026
ACM SIGGRAPH Asia	2026
ACM SIGGRAPH / Eurographics Symposium on Computer Animation	2026
ACM SIGGRAPH Motion, Interaction and Games	2026
IEEE Transactions on Visualization and Computer Graphics	2023, 2025, 2026
Eurographics	2026
Computer Graphics Forum	2026
Journal of Computational Physics	2025, 2026
Nuclear Engineering and Design	2025

GRANTS

Deutschlandstipendium

Selected based on academic performance and social commitment.

Germany

2019 – 2020

TEACHING

Seminars

Advanced Topics in Animation

University of Freiburg

2026

Supervised Graduate Students

Julian Karrer

University of Freiburg

2026

Joshua Fässler

Strong Rigid-Fluid Coupling using a Unified Implicit SPH Solver

University of Freiburg

2024 – 2025

Philipp Sandweg

Implicit SPH Simulation of Elastic Solids in a Multi-Material Framework

University of Freiburg

2023 – 2024

Andrés Sandoval

Global Matrix-free Pressure Solver for Performant SPH Fluid Simulations

University of Freiburg

2022 – 2023

Supervised Undergraduate Students

Johann-Frederik Isele

State Equation SPH Fluid Simulation

University of Freiburg

2026

Marius Hägele

Implementation and evaluation of a 2D state equation SPH fluid solver

University of Freiburg

2025 – 2026

Georgij Hergenröder

Smoothed Particle Hydrodynamics Fluid Simulation in 2D -

Implementation and Evaluation

University of Freiburg

2025 – 2026

SKILLS

Languages: Fluent in English and German (written, verbal, and reading)

Coding: C++, Python, C, C#, OpenCL, SYCL, CUDA

Frameworks: NumPy, Godot, Unreal Engine, Blender

